

Intramolecular Magnetic Exchange Coupling in Finite 7-Atom-Wide Armchair Graphene Nanoribbons: A Mean-Field Hubbard Study

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Abstract

We investigate the intramolecular magnetic exchange coupling J between the spin-polarized topological end states of finite 7-atom-wide armchair graphene nanoribbons (7-AGNRs) using two complementary approaches within the single-band Hubbard model: (i) a self-consistent mean-field Hubbard (MFH) calculation yielding $J = E_{\text{FM}} - E_{\text{AFM}}$ from total energies, and (ii) a projection onto an effective Hubbard dimer model giving $J = 4t_{\text{eff}}^2/U_{\text{eff}}$ via Anderson's superexchange formula. We systematically vary the ribbon length from $L = 4$ to $L = 20$ DBBA precursor monomers and demonstrate that J decays exponentially with a characteristic decay length $\xi \approx 0.38$ monomer units. Both methods agree quantitatively for $L \leq 11$; beyond this length, the MFH results saturate at the self-consistent-field noise floor ($\sim 10^{-10}$ meV), whereas the dimer model remains numerically robust.

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1 Introduction

Graphene nanoribbons (GNRs)—narrow strips of graphene with atomically defined edges—have emerged as a versatile platform for exploring quantum magnetism at the nanoscale. Unlike bulk graphene, which is diamagnetic and lacks intrinsic magnetic order, GNRs of specific widths and edge terminations can host robust spin-polarized states that are topologically protected and experimentally accessible through on-surface synthesis [11, 4]. Among the various GNR families, the **7-atom-wide armchair graphene nanoribbon (7-AGNR)** occupies a distinguished position. The 7-AGNR belongs to the $N = 3p + 1$ family of AGNRs (with $p = 2$), which exhibits a substantial band gap that makes it semiconducting. These ribbons are routinely synthesized by on-surface polymerization and cyclodehydrogenation of 10,10'-dibromo-9,9'-bianthryl (DBBA) precursor monomers on Au(111) [2]. Crucially, when a 7-AGNR is synthesized with finite length, its zigzag-terminated ends host **spin-polarized topological end states (SPTs)**—localized zero-mode-like states that carry a net magnetic moment of approximately $S = 1/2$ per end [1]. The topological origin of these states is encoded in a nontrivial $\mathbb{Z}_2$ invariant, which guarantees—through the bulk-boundary correspondence—the existence of in-gap boundary modes regardless of the microscopic details of the edge termination. The presence of two spatially separated spin- $1/2$ centers at the ends of a finite 7-AGNR gives rise to a natural question: *what is the effective magnetic coupling J between these end spins, and how does it depend on the ribbon length L ?* This coupling determines whether the ground state is an antiferromagnetic (AFM) open-shell singlet or a ferromagnetic (FM) triplet, and its magnitude—which decays exponentially with L —is directly relevant for potential applications in molecular spintronics and quantum information. In this work, we investigate the intramolecular exchange coupling J in finite 7-AGNRs using two complementary theoretical approaches within the framework of the single-band Hubbard model:

1. **The mean-field Hubbard (MFH) method**, in which we self-consistently solve the spin-resolved Hartree–Fock equations for both the AFM and FM magnetic configurations and extract $J = E_{\text{FM}} - E_{\text{AFM}}$ directly from the total energies.
2. **The effective Hubbard dimer model**, in which we project the full many-body problem onto a two-site model and compute $J = 4t_{\text{eff}}^2/U_{\text{eff}}$ using the Anderson superexchange formula.

We systematically vary the ribbon length from $L = 4$ to $L = 20$ DBBA precursor monomers, demonstrate that J decays exponentially with a characteristic decay length $\xi \approx 0.38$ monomer units, and compare the predictions of both methods to assess the range of validity and numerical limitations of each approach.

2 Theoretical Framework

2.1 Tight-Binding Description of Graphene Nanoribbons

The electronic structure of graphene and its nanostructures is well described at the single-particle level by a tight-binding (TB) Hamiltonian acting on the p_z orbitals of the carbon lattice. For graphene nanoribbons, it is essential to include hopping processes beyond nearest neighbors to correctly reproduce the width-dependent band-gap hierarchy [5, 10]. We employ a **third-nearest-neighbor** parameterization:

$$H_{\text{TB}} = - \sum_{\langle i,j \rangle} t_1 c_i^\dagger c_j - \sum_{\langle\langle i,j \rangle\rangle} t_2 c_i^\dagger c_j - \sum_{\langle\langle\langle i,j \rangle\rangle\rangle} t_3 c_i^\dagger c_j + \text{H.c.} \quad (1)$$

where $c_i^\dagger$ (c_i) creates (annihilates) an electron in the p_z orbital of site i , and the sums run over first-, second-, and third-nearest-neighbor pairs at distances $d_1 = a_{\text{CC}} = 1.42 \text{ \AA}$, $d_2 = \sqrt{3} a_{\text{CC}} \approx$

2.46 Å, and $d_3 = 2 a_{\text{CC}} = 2.84$ Å, respectively. The hopping parameters adopted throughout this work are:

$$t_1 = -2.7 \text{ eV}, \quad t_2 = -0.2 \text{ eV}, \quad t_3 = -0.18 \text{ eV}. \quad (2)$$

These values follow the parameterization of Hancock *et al.* [5] and are standard in the GNR literature. The inclusion of t_2 breaks particle-hole symmetry, while t_3 is critical for the correct ordering of the three AGNR band-gap families ($N = 3p, 3p + 1, 3p + 2$).

2.2 The Mean-Field Hubbard Model

To capture the effects of electron-electron interactions, we augment the tight-binding Hamiltonian with an on-site Hubbard repulsion term, leading to the single-band Hubbard model:

$$H = H_{\text{TB}} + U \sum_i n_{i\uparrow} n_{i\downarrow}, \quad (3)$$

where U is the on-site Coulomb repulsion and $n_{i\sigma} = c_{i\sigma}^\dagger c_{i\sigma}$ is the number operator for spin σ at site i . The Hubbard U for carbon p_z orbitals has been estimated at $U \approx 2.0\text{--}3.5$ eV in the literature [12, 13], depending on the level of theory and the degree of screening. We adopt $U = 3.0$ eV throughout this work, corresponding to $U/|t_1| \approx 1.11$. The exact solution of the Hubbard model is intractable for the system sizes considered here. We therefore resort to the **mean-field (Hartree–Fock) approximation**, in which the quartic interaction term is decoupled as:

$$U n_{i\uparrow} n_{i\downarrow} \approx U [\langle n_{i\uparrow} \rangle n_{i\downarrow} + n_{i\uparrow} \langle n_{i\downarrow} \rangle - \langle n_{i\uparrow} \rangle \langle n_{i\downarrow} \rangle]. \quad (4)$$

This replaces the many-body problem with an effective single-particle Hamiltonian that depends self-consistently on the spin-resolved site occupations $\{\langle n_{i\uparrow} \rangle, \langle n_{i\downarrow} \rangle\}$. The self-consistent field (SCF) equations are iterated until the density matrix converges to a tolerance of 10^{-13} , ensuring high numerical precision. A central output of the converged MFH calculation is the **local magnetic moment** at each site:

$$m_i = \langle n_{i\uparrow} \rangle - \langle n_{i\downarrow} \rangle, \quad (5)$$

which provides a spatially resolved map of the spin polarization across the nanoribbon. The total spin projection is $S_z = \frac{1}{2} \sum_i m_i$.

2.3 Topological Origin of the End States

The existence of spin-polarized end states in finite 7-AGNRs is not accidental but is guaranteed by the **topological classification** of the bulk band structure. Cao, Zhao and Louie [1] demonstrated that AGNRs can be classified by a $\mathbb{Z}_2$ topological invariant, in close analogy with the Su–Schrieffer–Heeger (SSH) model of polyacetylene. This invariant is directly related to the **Zak phase**—a Berry phase accumulated across the one-dimensional Brillouin zone:

$$\gamma = i \int_{\text{BZ}} \langle u_k | \nabla_k | u_k \rangle dk. \quad (6)$$

For systems with inversion symmetry, the Zak phase is quantized to $\gamma = 0$ or $\gamma = \pi$. The 7-AGNR, with appropriate zigzag termination, belongs to the topologically nontrivial class ($\gamma = \pi$, $\mathbb{Z}_2 = 1$). The **bulk-boundary correspondence** then mandates the existence of topologically protected boundary states—one at each end of the finite ribbon—that appear as in-gap modes with wave functions exponentially localized at the termini. This topological perspective is complemented by a chemical viewpoint rooted in **Clar’s sextet theory**: the aromatic pattern of fully resonant benzene rings within the AGNR determines whether the edge termination leaves unpaired radicals. Configurations that cannot be described by a “fully Clar” structure correspond to the topologically nontrivial phase and host the localized end states. In the non-interacting TB picture, the two end states hybridize through the exponentially decaying overlap

of their wave functions, forming the HOMO and LUMO of the finite ribbon. The hybridization gap $\Delta = \varepsilon_{\text{LUMO}} - \varepsilon_{\text{HOMO}}$ decreases exponentially with ribbon length. When electron-electron interactions are switched on (via U), each end state becomes singly occupied and acquires a net spin polarization, giving rise to the **emergent π -magnetism** that is characteristic of these systems.

2.4 Magnetic Configurations: AFM and FM States

The two end spins can couple in two distinct configurations:

- **AFM state (open-shell singlet):** The two end magnetic moments are antiparallel, $m_i > 0$ at one end and $m_i < 0$ at the other, with total $S_z = 0$. We emphasize that this is an *open-shell* singlet—unlike a closed-shell singlet, the local magnetic moments are nonzero; they merely cancel globally.
- **FM state (triplet):** The two end moments are parallel, with both ends showing the same sign of m_i and total $S_z = 1$. In the MFH calculation, this state is enforced by fixing the electron counts to $q = (N/2 + 1, N/2 - 1)$, where N is the total number of sites, thereby imposing one excess spin-up electron.

Lieb’s theorem [6] provides a rigorous constraint: for the repulsive Hubbard model at half-filling on a bipartite lattice, the ground-state total spin is $S = \frac{1}{2}|N_A - N_B|$, where N_A and N_B are the numbers of sites in the two sublattices. Since a pristine 7-AGNR has $N_A = N_B$ (balanced bipartite lattice), the ground state is guaranteed to be a singlet ($S = 0$), i.e., the AFM configuration.

2.5 The Effective Hubbard Dimer Model and Superexchange

The physics of the two coupled end spins can be elegantly captured by projecting the full many-electron problem onto a **two-site effective Hubbard dimer** [7]. This projection proceeds in three steps.

Step 1—Effective hopping t_{eff} . We diagonalize the non-interacting ($U = 0$) tight-binding Hamiltonian and identify the HOMO and LUMO, which are the bonding and antibonding combinations of the two end states. The effective inter-site hopping of the dimer is half their energy splitting:

$$t_{\text{eff}} = \frac{|\varepsilon_{\text{LUMO}} - \varepsilon_{\text{HOMO}}|}{2}. \quad (7)$$

Step 2—Localized end-state reconstruction. The HOMO and LUMO are delocalized (symmetric and antisymmetric) molecular orbitals. We recover the localized end-state wave function at, say, the left end via:

$$\psi_L(i) = \frac{\psi_{\text{HOMO}}(i) + \psi_{\text{LUMO}}(i)}{\sqrt{2}}. \quad (8)$$

Step 3—Effective Coulomb repulsion U_{eff} . The bare Hubbard U is projected onto the localized orbital through the **inverse participation ratio** of the wave function:

$$U_{\text{eff}} = U \sum_i |\psi_L(i)|^4. \quad (9)$$

This effective repulsion encodes the spatial extension of the end state: a highly localized state yields a large U_{eff} , while a delocalized one yields a small U_{eff} . In the **strong-correlation limit** ($t_{\text{eff}} \ll U_{\text{eff}}$), the half-filled Hubbard dimer maps onto an effective Heisenberg model

$H_{\text{eff}} = J \mathbf{S}_L \cdot \mathbf{S}_R$ with an antiferromagnetic exchange coupling given by Anderson's superexchange formula:

$$J = \frac{4t_{\text{eff}}^2}{U_{\text{eff}}} \quad (10)$$

Since $t_{\text{eff}} \propto e^{-L/\xi_0}$ (where ξ_0 is the decay length of the end-state wave function into the semi-conducting bulk), the exchange coupling also decays exponentially with ribbon length:

$$J(L) = A e^{-L/\xi}, \quad (11)$$

where $\xi = \xi_0/2$ reflects the fact that $J \propto t_{\text{eff}}^2$. The sign convention throughout this work is $J = E_{\text{FM}} - E_{\text{AFM}} > 0$ for an AFM ground state.

3 Computational Implementation

All calculations were performed using the `sisl` library [9] for geometry construction and tight-binding Hamiltonian setup, and the `hubbard` package (built on `sisl`) for the mean-field Hubbard self-consistent cycle.

3.1 Geometry Construction and Definition of L

A finite 7-AGNR of length L is constructed by tiling the crystallographic unit cell of a periodic 7-AGNR along the ribbon axis, followed by removal of periodic boundary conditions and iterative pruning of dangling bonds (coordination-1 carbon atoms).

Definition of L . The `sisl` function `agnr(width=7)` generates an orthogonal unit cell containing 14 carbon atoms with a lattice constant $a = 3a_{\text{CC}} = 4.26 \text{ \AA}$ along the ribbon axis. In our code, each length unit L corresponds to a tile of 2 such unit cells (28 atoms, spanning $8.52 \text{ \AA}$). Since each DBBA precursor monomer (10,10'-dibromo-9,9'-bianthryl) contributes exactly 28 carbon atoms to the ribbon upon cyclodehydrogenation, L is numerically identical to the **number of DBBA precursor monomers** incorporated into the finite ribbon.

```

1 import sisl
2 def build_7agnr_geometry(L):
3     """
4     Build a finite 7-AGNR geometry with L monomer units.
5     1 monomer = 2 unit cells. Iterative pruning of dangling bonds.
6     """
7     uc = sisl.geom.agnr(width=7, bond=1.42)
8     geom = uc.tile(2 * L, axis=0)
9     geom.set_nsc([1, 1, 1]) # Remove periodicity -> finite molecule
10    while True:
11        to_remove = []
12        for ia in geom:
13            idx = geom.close(ia, R=1.5)
14            if len(idx) == 2: # self + 1 neighbour = coordination 1
15                to_remove.append(ia)
16        if not to_remove:
17            break
18        geom = geom.remove(to_remove)
19    return geom

```

Listing 1: Geometry construction of a finite 7-AGNR with L DBBA monomer units.

The function `sisl.geom.agnr(width=7, bond=1.42)` generates the unit cell of a 7-AGNR with C–C bond length $a_{CC} = 1.42 \text{ \AA}$. The `tile(2*L, axis=0)` call replicates this cell $2L$ times along the ribbon direction. Setting the supercell to `[1, 1, 1]` removes all periodicity, yielding a finite molecular cluster. The `while` loop then iteratively identifies and removes atoms with only one neighbor (dangling bonds at the ends), continuing until no further undercoordinated atoms remain. This ensures physically meaningful zigzag-terminated edges.

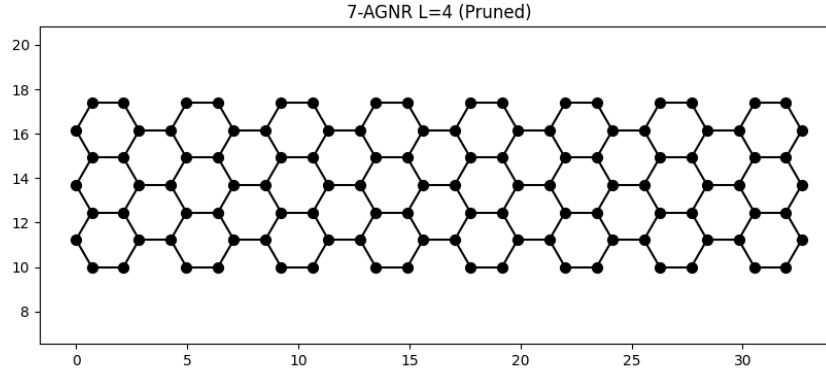


Figure 1: Atomic structure of a finite 7-AGNR with $L = 4$ monomer units after iterative pruning of dangling bonds. The ribbon exhibits clean zigzag terminations at both ends, which host the topological end states upon inclusion of electron-electron interactions.

3.2 Mean-Field Hubbard Self-Consistent Calculation

With the geometry defined, we construct a spin-polarized 3NN tight-binding Hamiltonian using the `sp2` function from the `hubbard` package. We then solve the MFH equations for both the AFM and FM configurations. The edge atoms at each end of the ribbon are identified geometrically by their x -coordinate:

```

1 from hubbard import sp2, HubbardHamiltonian, density
2 def calculate_J_mfh(geom, U_val=3.0, t1=-2.7, t2=-0.2, t3=-0.18):
3     """
4     Compute J = E_FM - E_AFM from the self-consistent MFH cycle.
5     """
6     H_tb = sp2(geom, t1=t1, t2=t2, t3=t3, spin='polarized')
7     # Identify zigzag-edge atoms at the two ends
8     xyz = geom.xyz
9     x_coords = xyz[:, 0]
10    tol_edge = 2.0
11    left_atoms = np.where(x_coords < x_coords.min() + tol_edge)[0]
12    right_atoms = np.where(x_coords > x_coords.max() - tol_edge)[0]
13    # -- AFM state (open-shell singlet) --
14    HH_afm = HubbardHamiltonian(H_tb, U=U_val, kT=1e-6, nkpt=[1,1,1])
15    HH_afm.set_polarization(left_atoms, dn=right_atoms)
16    HH_afm.converge(density.calc_n, tol=1e-13, print_info=False)
17    E_afm = HH_afm.Etot
18    # -- FM state (triplet, Sz = 1) --
19    N_tot = geom.na
20    q_fm = (N_tot // 2 + 1, N_tot // 2 - 1)
21    HH_fm = HubbardHamiltonian(H_tb, U=U_val, kT=1e-6, nkpt=[1,1,1],
22                               q=q_fm)
23    all_edge = np.concatenate((left_atoms, right_atoms))
24    HH_fm.set_polarization(all_edge)
25    HH_fm.converge(density.calc_n, tol=1e-13, print_info=False)

```

```

26 E_fm = HH_fm.Etot
27 return {'J': E_fm - E_afm, 'E_afm': E_afm, 'E_fm': E_fm}

```

Listing 2: MFH self-consistent calculation for AFM and FM states.

Several computational choices deserve comment:

- **Spin initialization.** For the AFM state, we seed the SCF cycle with spin-up polarization on the left-edge atoms and spin-down on the right-edge atoms. For the FM state, all edge atoms are initialized with spin-up polarization. This ensures convergence to the desired magnetic configuration.
- **Electron count.** The AFM state has equal numbers of spin-up and spin-down electrons ($N_\uparrow = N_\downarrow = N/2$). The FM state is enforced by setting $q = (N/2 + 1, N/2 - 1)$, which fixes $S_z = 1$ and prevents the SCF cycle from collapsing to the lower-energy singlet.
- **Electronic temperature.** We use an extremely low broadening $k_B T = 10^{-6}$ eV to ensure sharp Fermi occupations, appropriate for an insulating system at zero temperature.
- **Convergence tolerance.** A tolerance of 10^{-13} on the density matrix provides the numerical precision required to resolve the minute energy differences $J = E_{\text{FM}} - E_{\text{AFM}}$, which can be as small as 10^{-9} meV for long ribbons.

3.3 Effective Hubbard Dimer Projection

The second method bypasses the self-consistent cycle entirely and computes J analytically from the non-interacting TB spectrum:

```

1 def calculate_J_dimer(geom, U_bare=3.0, t1=-2.7, t2=-0.2, t3=-0.18):
2     """
3     Project the 7-AGNR physics onto an effective Hubbard dimer.
4     t_eff = Delta_gap / 2
5     U_eff = U_bare * sum_i |psi_L(i)|^4
6     J      = 4 t_eff^2 / U_eff
7     """
8     H_tb = sp2(geom, t1=t1, t2=t2, t3=t3, spin='unpolarized')
9     # Full diagonalisation
10    evals, evecs = H_tb.eigh(eigvals_only=False)
11    N = len(evals)
12    idx_homo = N // 2 - 1
13    idx_lumo = N // 2
14    # t_eff = half the hybridisation gap
15    t_eff = np.abs(evals[idx_lumo] - evals[idx_homo]) / 2.0
16    psi_homo = evecs[:, idx_homo]
17    psi_lumo = evecs[:, idx_lumo]
18    # Reconstruct the localised end state
19    psi_L = (psi_homo + psi_lumo) / np.sqrt(2)
20    U_eff = U_bare * np.sum(np.abs(psi_L)**4)
21    J_dimer = 4.0 * t_eff**2 / U_eff
22    return {'J': J_dimer, 't_eff': t_eff, 'U_eff': U_eff}

```

Listing 3: Effective Hubbard dimer projection.

This function diagonalizes the unpolarized 3NN Hamiltonian to obtain the full eigenspectrum $\{\varepsilon_n, \psi_n\}$. The HOMO and LUMO—identified as the states straddling the Fermi level—are the bonding and antibonding combinations of the two topological end states. The effective hopping t_{eff} is extracted as half the HOMO–LUMO gap, the localized end-state orbital ψ_L is reconstructed by superposition (Eq. 8), and the effective on-site repulsion U_{eff} is computed via the inverse participation ratio (Eq. 9). Finally, the superexchange formula (Eq. 10) yields the exchange coupling.

3.4 Spin Polarization Diagnostics

To verify that the SCF cycle converges to the correct magnetic state, we generate spatially resolved maps of the local spin polarization $m_i = n_{i\uparrow} - n_{i\downarrow}$ for both the AFM and FM configurations. Each atom is rendered at its crystallographic position, colored according to its local moment m_i using a diverging RdBu_r colormap: red indicates spin-up polarization ($m_i > 0$), blue indicates spin-down ($m_i < 0$), and white indicates no net polarization. Both panels share the same color scale to enable direct visual comparison between the AFM and FM states.

3.5 Length Sweep and Exponential Fit

The main computational loop sweeps over ribbon lengths $L = 4$ to $L = 20$ monomer units, computing J by both methods at each length. The exponential decay $J(L) = A e^{-L/\xi}$ is extracted via a log-linear fit to the dimer data:

```

1 # Exponential fit:  $J = A * \exp(-L / \xi)$ 
2 log_J = np.log(J_dimer_meV)
3 coeffs = np.polyfit(L_arr, log_J, 1) # slope =  $-1/\xi$ 
4 xi = -1.0 / coeffs[0] # decay length
5 A_fit = np.exp(coeffs[1])

```

Listing 4: Exponential fit of the exchange coupling decay.

This procedure exploits the linearity of $\ln J$ vs. L to obtain both the amplitude A and the decay length ξ from a simple polynomial regression.

4 Results and Discussion

4.1 Spin Polarization Maps

We begin by examining the spatial distribution of the local spin polarization in the AFM and FM states for a representative ribbon of length $L = 6$ monomer units.

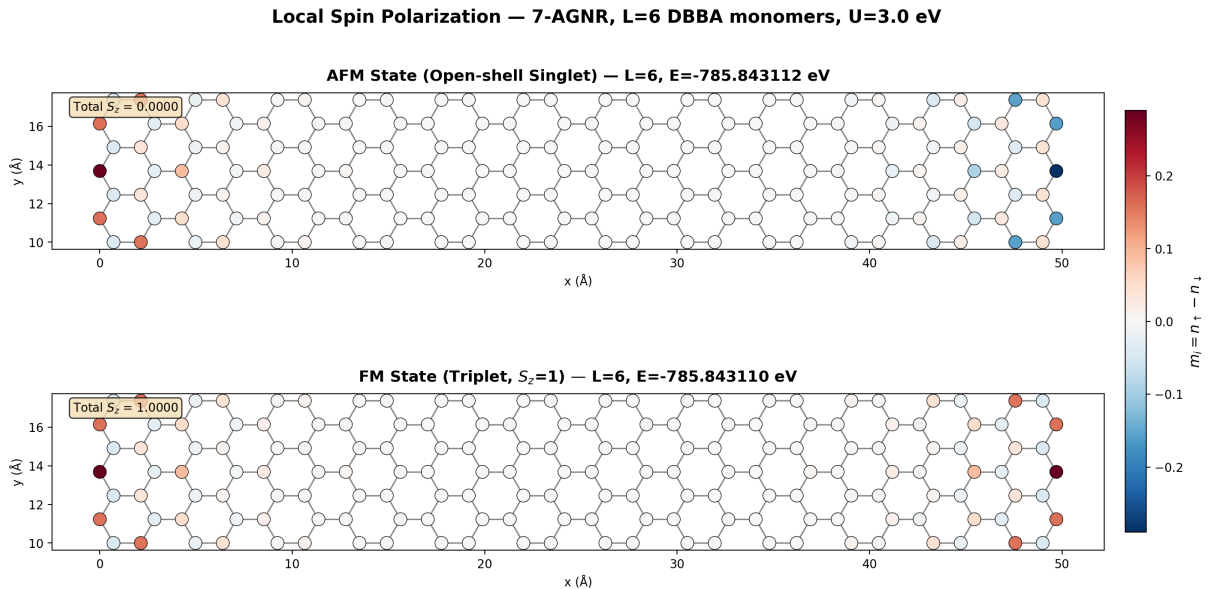


Figure 2: Local spin polarization $m_i = n_{i\uparrow} - n_{i\downarrow}$ for a finite 7-AGNR with $L = 6$ monomer units and $U = 3.0$ eV. **Top:** AFM state (open-shell singlet) with $E_{\text{AFM}} = -785.843112$ eV and total $S_z = 0.0000$. **Bottom:** FM state (triplet) with $E_{\text{FM}} = -785.843110$ eV and total $S_z = 1.0000$. The color scale maps m_i from spin-down (blue) to spin-up (red).

Several important features are immediately apparent from Fig. 2:

1. **Edge localization.** In both states, the spin polarization is strongly concentrated at the zigzag-terminated ends of the ribbon, with the interior atoms showing negligible magnetization. This confirms the topological nature of the end states—the spin density tracks the localized zero-mode wave function.
2. **AFM pattern.** In the AFM state (top panel), the left end shows predominantly spin-up polarization (red), while the right end shows spin-down polarization (blue), with the two patterns related by the combined symmetry of spatial inversion and spin reversal. The total $S_z = 0$ confirms the singlet character.
3. **FM pattern.** In the FM state (bottom panel), both ends display the same spin-up polarization (red), with a total $S_z = 1$ as expected for a triplet state.
4. **Energy difference.** The energy splitting between the two states is $J = E_{\text{FM}} - E_{\text{AFM}} = 2 \times 10^{-6} \text{ eV} \approx 2 \text{ } \mu\text{eV}$. This remarkably small value reflects the weak hybridization between the two end states at this ribbon length, and underscores the numerical challenge of resolving J for longer ribbons.

4.2 Exchange Coupling from the Effective Hubbard Dimer

Figure 3 presents the exchange coupling J computed via the effective Hubbard dimer model as a function of ribbon length L , displayed on a semilogarithmic scale.

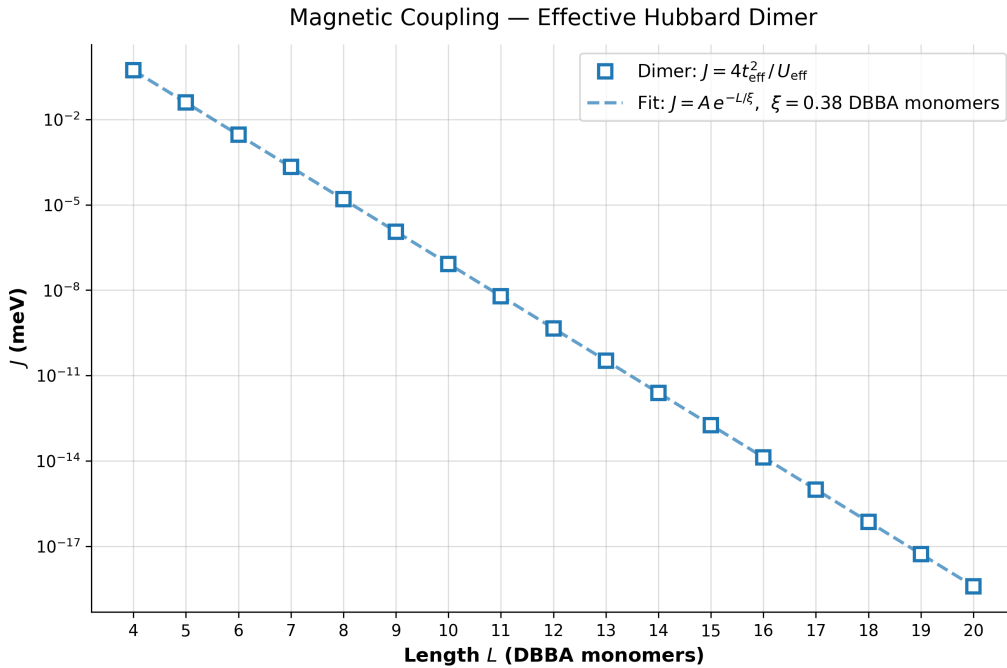


Figure 3: Exchange coupling J computed from the effective Hubbard dimer formula $J = 4t_{\text{eff}}^2/U_{\text{eff}}$ as a function of ribbon length L (in DBBA monomer units), with $U = 3.0 \text{ eV}$. Blue squares: computed data points; dashed line: exponential fit $J = A e^{-L/\xi}$ yielding a decay length $\xi = 0.38$ monomer units.

The dimer model yields strikingly clean results: the data points fall on a perfect straight line in the $\ln J$ vs. L representation, confirming the expected exponential decay over **more than 20 orders of magnitude**—from $J \sim 0.1 \text{ meV}$ at $L = 4$ down to $J \sim 10^{-19} \text{ meV}$ at $L = 20$. The fitted decay length is $\xi = 0.38$ monomer units, indicating extremely rapid attenuation of

the magnetic coupling with increasing ribbon length. This rapid decay is a direct consequence of the sizable band gap of the 7-AGNR (~ 1.5 eV at the 3NN TB level), which governs the exponential decay of the end-state wave functions into the bulk. The decay length of t_{eff} (and hence of the wave-function overlap) is $\xi_0 = 2\xi \approx 0.76$ monomer units, consistent with the large gap-to-bandwidth ratio of this system.

4.3 Exchange Coupling from the Mean-Field Hubbard Method

Figure 4 shows the exchange coupling computed self-consistently via the MFH method.

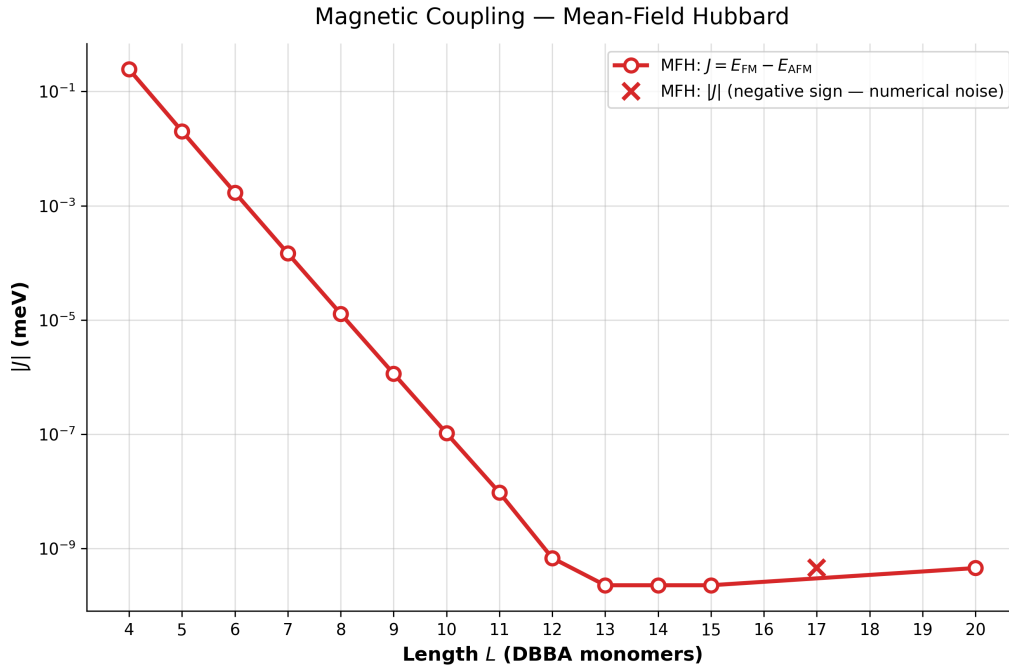


Figure 4: Exchange coupling $|J|$ extracted from the MFH total energies ($J = E_{\text{FM}} - E_{\text{AFM}}$) as a function of ribbon length L . Open circles: positive J (physical AFM ground state); the cross at $L = 17$ marks a negative J (sign flip attributed to numerical noise). The data exhibit clean exponential decay for $L \leq 11$, followed by saturation at the SCF noise floor for $L \geq 12$.

The MFH results reveal two distinct regimes:

1. **Physical regime** ($L = 4\text{--}11$). The computed J decays exponentially, in excellent quantitative agreement with the dimer model (see Sec. 4.4). The values span from $J \approx 0.2$ meV at $L = 4$ to $J \approx 10^{-8}$ meV at $L = 11$.
2. **Noise-dominated regime** ($L \geq 12$). The computed J saturates at a plateau of approximately $|J| \sim 10^{-10}$ meV, no longer tracking the true (exponentially decaying) physical coupling. At $L = 17$, the sign of J flips to negative—a clear signature that the energy difference has dropped below the intrinsic numerical precision of the SCF cycle. This defines the **SCF noise floor**, which sets a lower bound on the smallest resolvable J for a given convergence tolerance.

4.4 Comparison of the Two Methods

Figure 5 overlays the results of both methods on a common logarithmic scale, providing a direct assessment of their agreement and respective limitations.

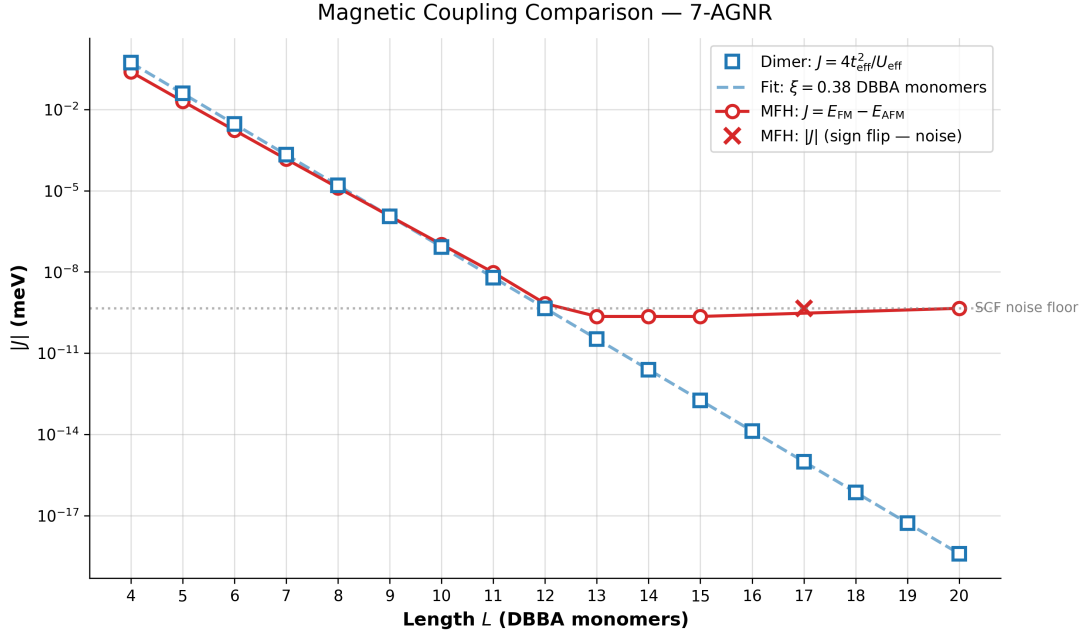


Figure 5: Comparison of the exchange coupling $|J|$ as a function of ribbon length L for the two computational methods. Blue squares: effective Hubbard dimer ($J = 4t_{\text{eff}}^2/U_{\text{eff}}$) with exponential fit (dashed line, $\xi = 0.38$ monomer units). Red circles: MFH ($J = E_{\text{FM}} - E_{\text{AFM}}$); the red cross marks a sign-flipped data point. The horizontal dotted line marks the SCF noise floor.

The comparison yields three key conclusions:

1. **Quantitative agreement.** For ribbon lengths $L = 4$ to $L \approx 11$, the two methods agree quantitatively, validating the Hubbard dimer projection as a faithful low-energy description of the magnetic physics.
2. **Breakdown of the MFH method.** Beyond $L \approx 12$, the MFH method can no longer resolve the exponentially small energy splitting J , and the extracted values are dominated by SCF numerical noise. The noise floor of $\sim 10^{-10}$ meV corresponds to a relative energy precision of $\sim 10^{-16}$ in the total energy (which is of order $\sim 10^3$ eV for these systems), consistent with the limits of double-precision floating-point arithmetic.
3. **Robustness of the dimer model.** The effective Hubbard dimer model, which relies on the diagonalization of the non-interacting Hamiltonian, does not suffer from this numerical limitation and can predict J to arbitrarily small values. This makes it the method of choice for long ribbons or for systematic parametric studies where the MFH method would be prohibitively expensive or numerically unreliable.

We note that the crossover between the two regimes occurs at $L \approx 12$ monomer units, corresponding to a physical coupling of $J \sim 10^{-9}$ meV $\sim 10^{-5}$ μ eV. Below this scale, only the dimer model provides meaningful predictions.

5 Conclusions

We have presented a systematic computational study of the intramolecular magnetic exchange coupling J between the topological end states of finite 7-AGNRs, employing two complementary approaches within the mean-field Hubbard framework. Our principal findings are:

1. **Spin-polarized end states are topologically robust.** The local spin polarization maps confirm that finite 7-AGNRs host two well-defined spin-1/2 centers at their zigzag-terminated ends, consistent with the $\mathbb{Z}_2 = 1$ topological classification of this ribbon. The AFM (open-shell singlet) state is the ground state, in agreement with Lieb’s theorem for balanced bipartite lattices.
2. **The exchange coupling decays exponentially with ribbon length.** Both computational methods consistently predict $J(L) = A e^{-L/\xi}$ with a decay length $\xi \approx 0.38$ monomer units. This rapid decay reflects the large band gap of the 7-AGNR, which exponentially suppresses the hybridization between the two end-state wave functions.
3. **The effective Hubbard dimer model faithfully captures the low-energy physics.** The superexchange formula $J = 4t_{\text{eff}}^2/U_{\text{eff}}$ reproduces the self-consistent MFH results within the physical regime ($L \leq 11$), validating the projection of the full many-body problem onto a two-site model.
4. **The MFH method has an intrinsic numerical precision limit.** For $L \geq 12$, the exponentially small J drops below the SCF noise floor ($\sim 10^{-10}$ meV), and the MFH results are no longer reliable. The dimer model does not suffer from this limitation and provides a computationally efficient and numerically robust alternative for long ribbons.

These results establish a solid methodological foundation for the study of length-dependent magnetic interactions in GNR-based systems and open the door to the design of molecular spin chains with tunable exchange couplings.

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